PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

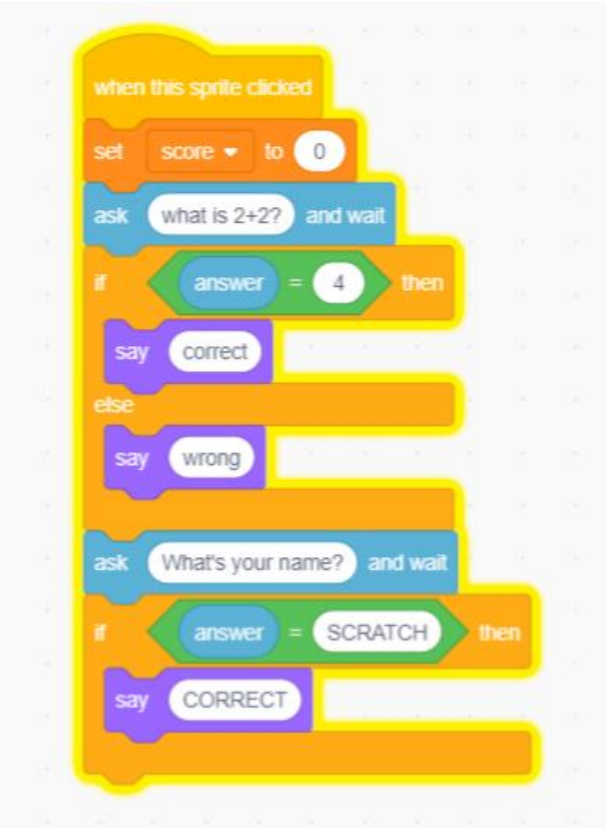
Programming Fundamentals Lab 01	**Instructor: Mr. Muhammad Hamza**
Program: AI- Fall 2025	**Date:10/15/2025**
Marks: 10	**Lab Engineer: MS. Qandil Fatima**
Student Name: ESHAL AYUB	**Reg. No: B25F1061AI100**
Instructions: • Submit the file with your names following your registration numbers like **AI001_Name.** • **Copy Paste work will be given no marks.**	

Lab Task

1. Create a simple quiz game (minimum of 2 questions). Provide the final code snap along with the results. Also write the algorithm of the program.

Algorithm:

1. Start the program when the sprite is clicked.
2. Set the score to 0.
3. Ask the first question ("What is 2 + 2?").
4. If the answer is correct (4), say "Correct!" and increase score by 1. Otherwise, say "Wrong!".
5. Ask the second question ("WHAT'S YOUR NAME"?
6. If the answer is correct ("SCRATCH"), say "Correct!" and increase score by 1. Otherwise, say "Wrong!".
7. Display the final score.
8. End the program.



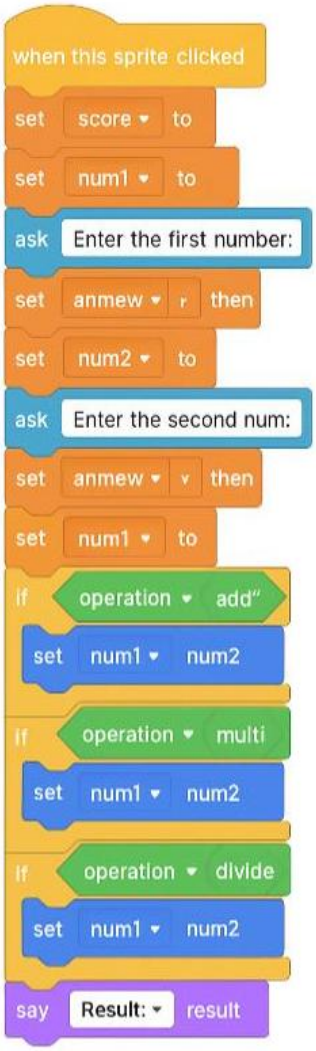
Result:

The Scratch quiz game successfully asks two questions, checks the user's answers, and displays the total score at the end.

2. Write a scratch program to add, subtract, multiply, and divide 2 numbers (a simple calculator)? (Insert the code snap along with output and write its algorithm).

Algorithm:

1. Start the program when the sprite is clicked.
2. Ask the user to enter the first number and store it in variable num1.
3. Ask the user to enter the second number and store it in variable num2.
4. Ask the user to choose an operation (add, subtract, multiply, or divide) and store it in variable operation.
5. If the operation is added, set result = num1 + num2.
6. If the operation is subtracted, set result = num1 - num2.
7. If the operation is multiply, set result = num1 * num2.
8. If the operation is divided, set result = num1 / num2.
9. Display the result using a say block.
10. Stop the program.



Result:

The Scratch calculator program successfully performs basic arithmetic operations addition, subtraction, multiplication, and division between two user-input numbers. It demonstrates the use of variables, conditional statements, and user interaction blocks in Scratch, showing how simple logic can be used to create an interactive calculator.